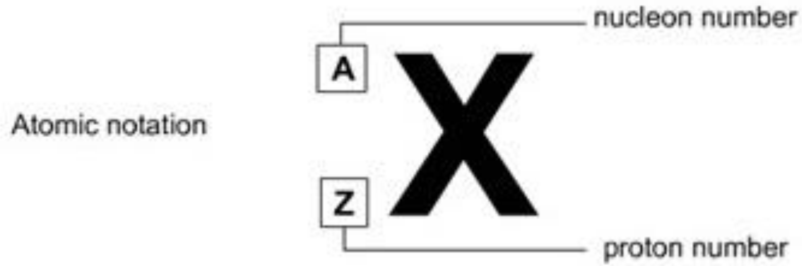
Topic : Nuclear Physics

- Mass excess and nuclear binding energy
- Radioactive decay

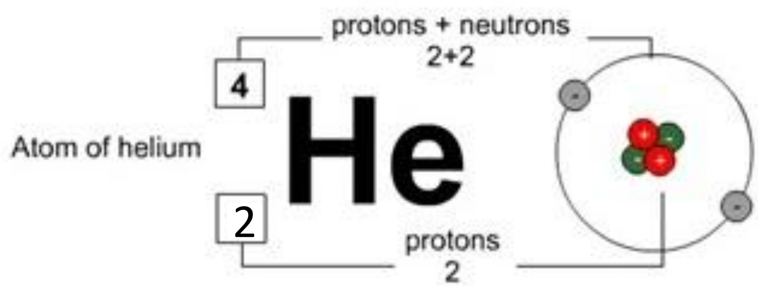
To describe the number of particles in a given atom, we use this notation:



The top number (A) is called the **nucleon number** (as it is the number of things in the nucleus of the atom) or the **mass number** (as it is the mass of the atom.)

The bottom number (Z) is called the **proton number** (as it is the number of protons) or the **atomic number** (as it is the number that tells you which element the atoms belongs to).

The letters give you a clue as to the name of the element. ***For example here is an atom of helium:***.



Recap…

distinguish between nucleon number and proton number

Nucleon number A:	The number of nucleons (Protons and Neutrons)
Proton number Z:	The number of protons in the nucleus
Neutron number N:	The number of neutrons in the nucleus

show an understanding that an element can exist in various isotopic forms, each with a different number of neutrons

Nuclide:	An atom with a particular number of protons and neutrons
Isotope:	Isotopes are nuclides that contain the same number of protons, but different numbers of neutrons.
Nucleon:	Component of the nucleus = Protons and Neutrons

Decay equations

To show what happens before and after a nuclear reaction (reaction involving the nucleus of an atom) we use equations that show both the proton (Z) and nucleon number (A). To balance a nuclear equation (left side and right side) you have to make sure that the sum of the nucleon (top) numbers on the left hand side equals the sum of the nucleon numbers on the right hand side AND the sum of the proton numbers on both sides also balance.

For Example:.

$$^{14}_{7}N + {}^{4}_{2}He \rightarrow {}^{17}_{8}O + {}^{1}_{1}H$$

Note : This equation is an example for balanced equation. Do not consider it for alpha emission.

Top row = 18 on both the left and right sides.

Bottom row = 9 on both the left and right sides.

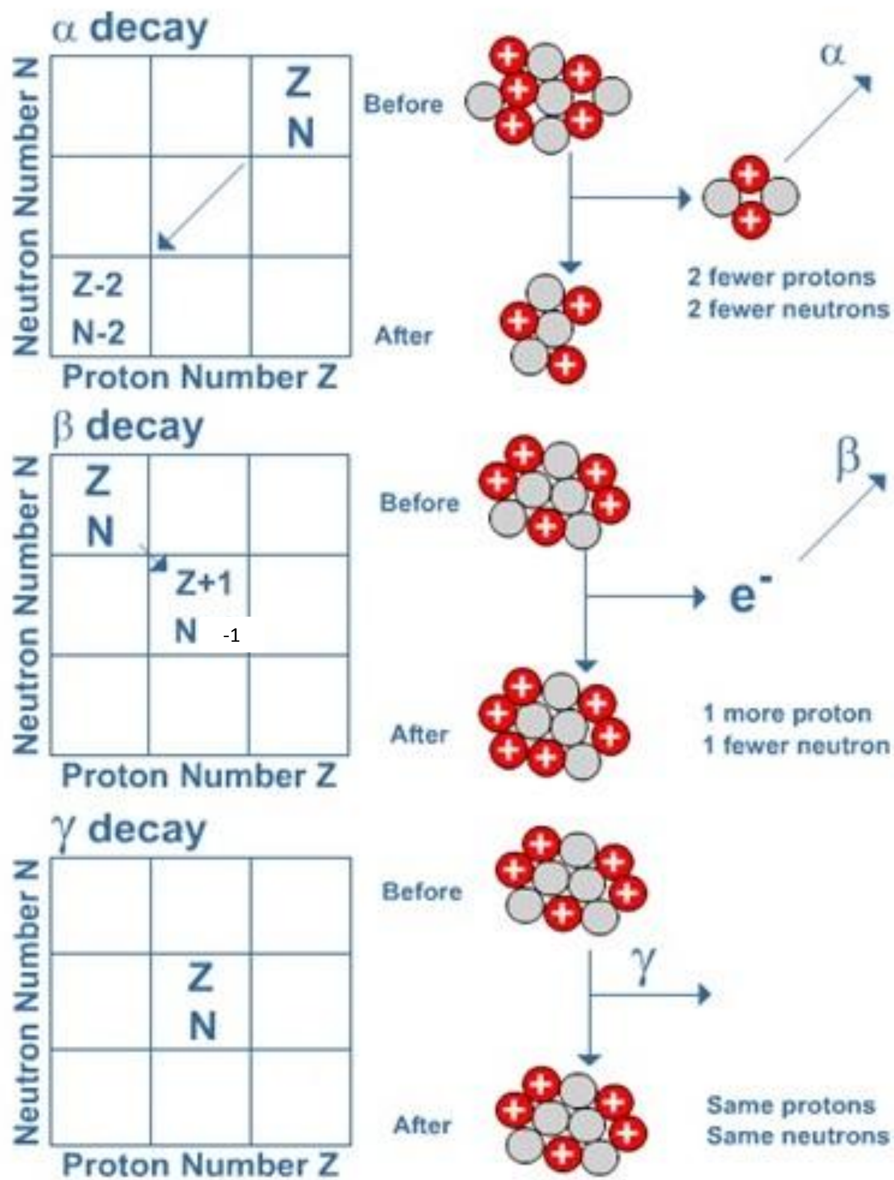
This is a balanced equation.

Nuclear equations such as these are useful for explaining what happens in radioactive decay processes.

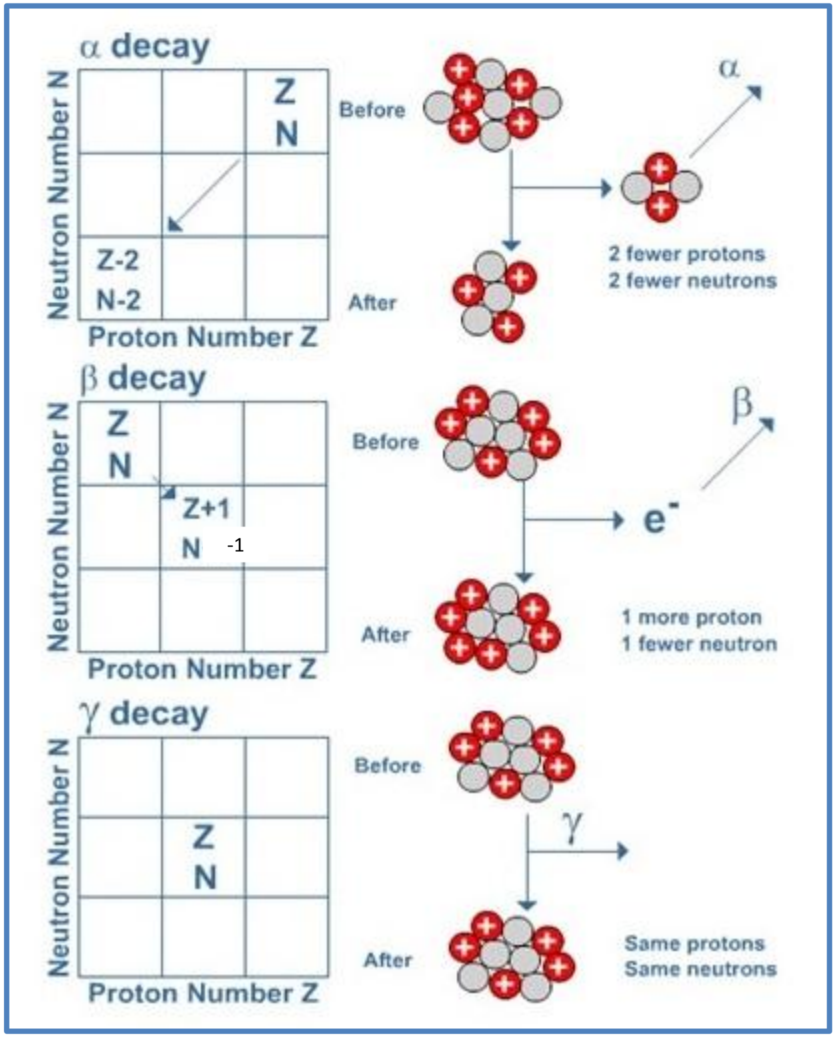
Unstable nuclei emit **alpha, beta** or **gamma** radiation in order to become more stable.

As a result of emitting this radiation the character of the nucleus remaining is changed. This is **radioactive decay**.

The diagrams below show what happens when a nucleus emits alpha, beta or gamma radiation.



- In alpha decay 2 protons and 2 neutrons are emitted. Notice that this reduces the nucleon number by 4 and the proton number by 2. A new element is thus formed.
- In beta decay a neutron changes into a proton (which remains in the nucleus) and an electron (which is emitted as beta radiation). The net effect is an increase in proton number by 1, while the nucleon number stays the same. Again a new element is formed.
- When a nucleus has undergone alpha or beta decay it is often left in a high-energy (**excited**) state. This energy can be lost in the form of an emitted a gamma ray. Because the composition of the nucleus is unchanged no new element is formed.



Recap…

represent simple nuclear reactions by nuclear equations of the form

$^{14}{}_{7}N + {}^{4}{}_{2}He$ --- $^{17}{}_{8}O + {}^{1}{}_{1}H$

Alpha decay….

$${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$$

Example….

$${}^{241}_{95}Am \rightarrow {}^{237}_{93}Np + {}^{4}_{2}\alpha$$

Beta decay

$${}^{A}_{Z}X \rightarrow {}_{Z+1}^{A}Y + {}^{0}_{-1}\beta + \bar{\nu}$$

$${}^{90}_{38}Sr \rightarrow {}^{90}_{39}Y + {}^{0}_{-1}\beta + \bar{\nu}$$

Gamma decay

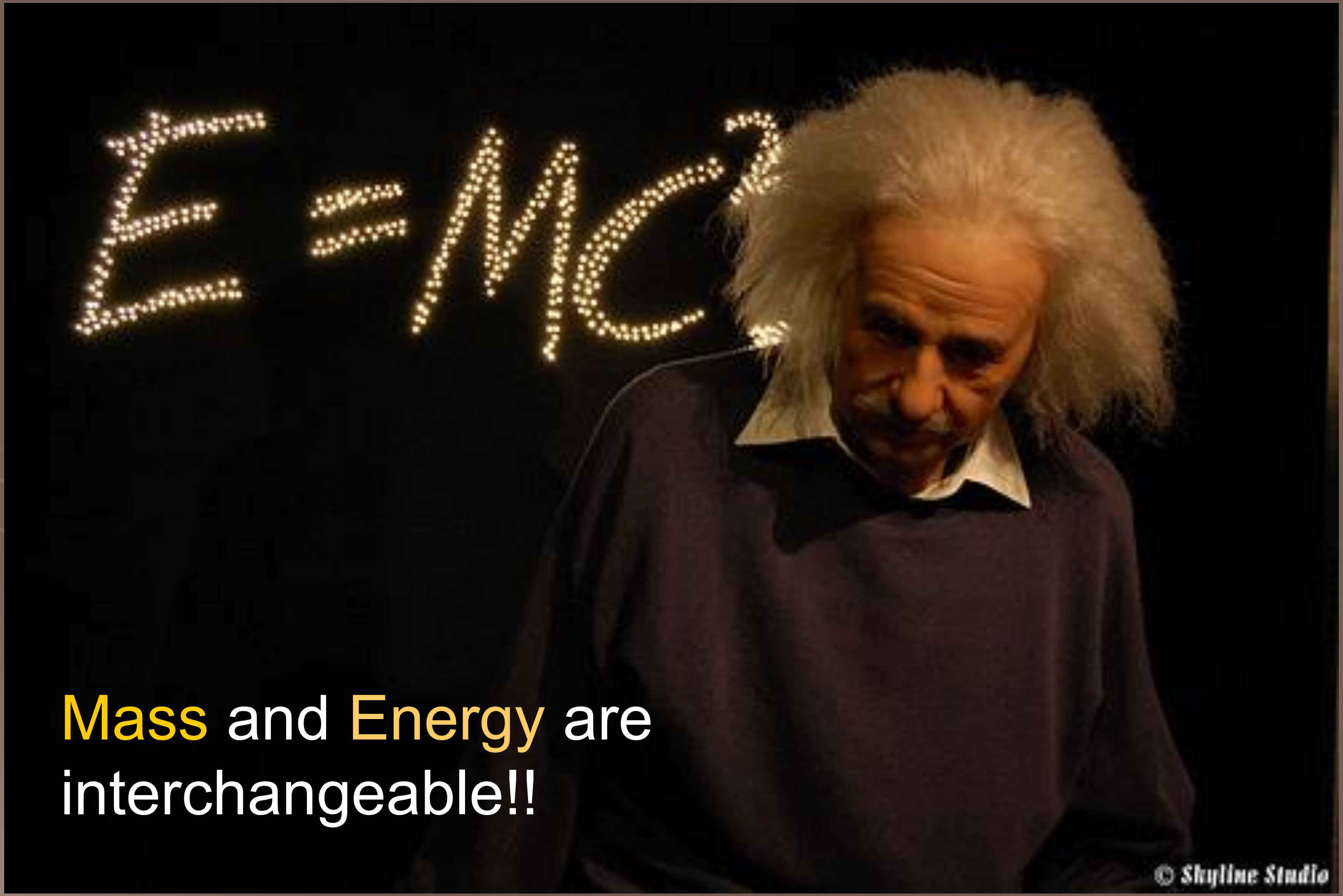
$${}^{A}_{Z}X^{*} \rightarrow {}^{A}_{Z}X + {}^{0}_{0}\gamma$$

In any nuclear reaction the following must *always* be true:

- The total atomic number before the reaction must be the same as the total atomic number after the reaction.
- The total atomic mass before the reaction must be the same as the total atomic mass after the reaction.

The first requirement above is the statement of the conservation of charge in nuclear reactions. The second requirement is the statement of the conservation of nucleon number.

Einstein’s Famous Equation

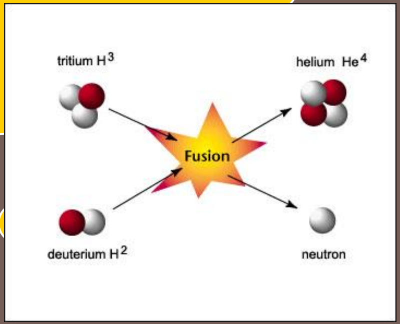


Mass and Energy are interchangeable!!

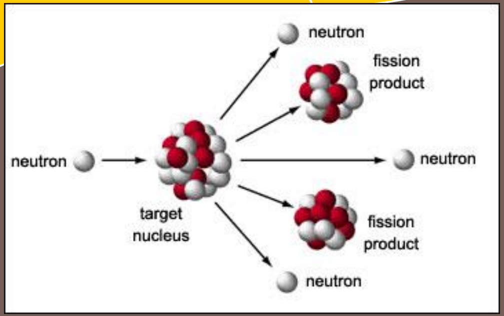
Where & When??

It happens only at the **nuclear level**

When two nuclei combine (Fusion)



OR When a nucleus breaks up (Fission)



Mass Excess

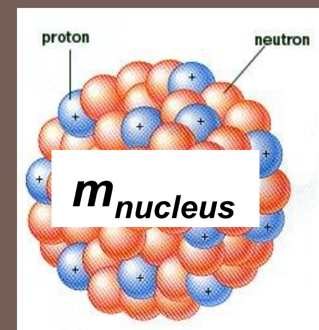
Proton

m_p

Neutron

m_n

Nucleus



When protons and neutrons come together to form a nucleus, the mass of the nucleus is less than the sum of the masses of separated protons and neutrons.

This difference in mass is called the mass excess or mass defect of the nucleus.

Mass Excess (Defect)

= Mass of separated protons & neutrons – mass of nucleus

$$\Delta m = Zm_p + Nm_p - m_{nucleus}$$

Where Z is the number of proton and N the number of neutrons in the nucleus.

Atomic Mass Unit (u)

- At nuclear level, the masses of the nuclei and nucleons are so small that the unit kg is too big and clumsy to be used.
- Instead the atomic mass unit (u) is used.
- One atomic mass unit (1 u) is defined as being equal to one-twelfth the mass of a carbon-12 atom.
- 1 u = 1.66×10^{-27} kg
- Using this scale of measurement, to six decimal places, we have

  proton mass, $m_p = 1.007276$ u

  neutron mass, $m_n = 1.008665$ u

  electron mass, $m_e = 0.000549$ u

Example 1

Calculate the mass defect for a carbon-14 nucleus $\left({}^{14}_{6}\mathrm{C}\right)$

The measured mass is 14.003240 u.

Calculate also the energy-equivalent of this mass loss in eV.

Solution:

Carbon-14 has 6 protons and 8 neutrons

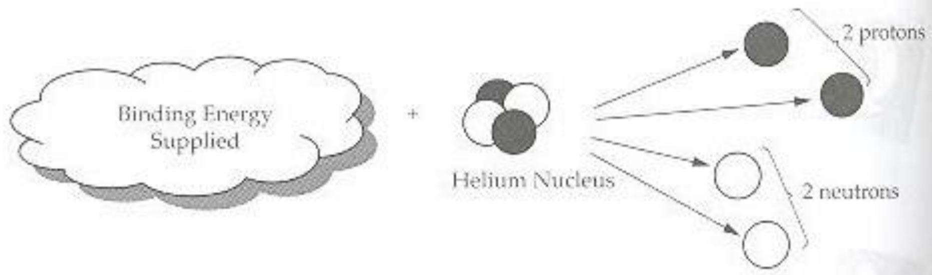
Mass defect = 6 (1.007276) + 8 (1.008665) –14.003240

= 0.109736 u

$E = mc^2 = [(0.109736 \times 1.66 \times 10^{-27}) \times (3.00 \times 10^{8})^2] / (1.6 \times 10^{-19})$

$= 102$ MeV

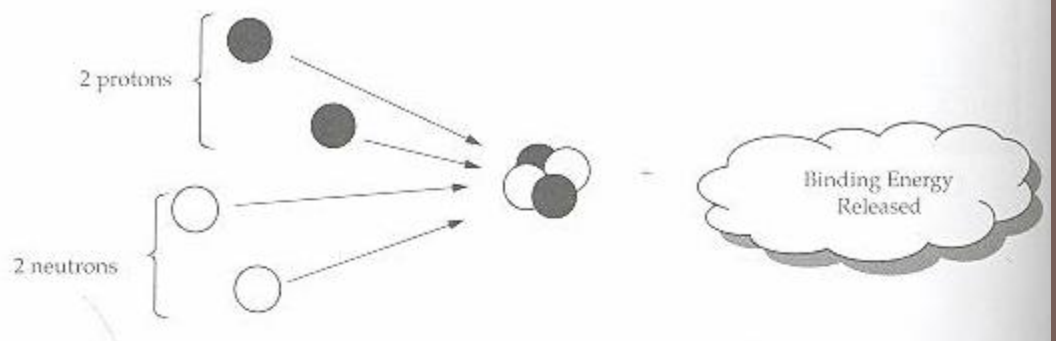
Binding energy



▲ Fig. 20.3 Energy is supplied to separate the nucleons

- Within the nucleus, there are strong forces which bind the protons and neutrons together.
- To completely separate all these nucleons requires energy.
- This energy is referred to as the binding energy.
- Binding energy is defined as the energy required to completely separate all nucleons of a nucleus.
- It is the energy equivalent of the mass defect of a nucleus.

Binding Energy



▲ Fig. 20.4 Separated nucleons combine to form helium nucleus

- When separated nucleons combine to form a nucleus, there is a reduction of mass and an equivalent amount of binding energy is released.

Binding Energy Per Nucleon

Binding energy per nucleon is the total binding energy of the nucleus divided by the total number of nucleons.

Binding energy per nucleon

$$= \frac{\text{Binding energy of the nucleus}}{\text{Nucleon number of the nucleus}} \text{ J per nucleon}$$

It is a measure of the stability of the nucleus.

Example 2

Helium nucleus is represented by ^4_2He. Use the information given below to calculate the

(a) mass defect,

(b) binding energy and

(c) binding energy per nucleon of the helium nucleus.

Mass of proton = 1.00728 u

Mass of neutron = 1.00866 u

Mass of helium nucleus = 4.001504 u

Solution 2

$2p + 2n \rightarrow {}^{4}_{2}He$ + binding energy

(a) Mass of separate constituent nucleons $= 2(1.00728) + 2(1.00866)$
$= 4.03188$ u

Mass of helium nucleus $= 4.001504$ u

mass defect = mass of separated nucleons − mass of nucleus

$$\Delta m = 4.03188 \text{ u} - 4.001504 \text{ u}$$
$$= 0.030376 \text{ u}$$

(b) Binding energy $= \Delta m \times c^2$

$$= 0.030376 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$$
$$= 4.5381744 \times 10^{-12} \text{ J}$$
$$= \frac{4.5381744 \times 10^{-12}}{1.60 \times 10^{-19}} \text{eV}$$
$$= 2.84 \times 10^7 \text{ eV}$$
$$= 28.4 \text{ MeV}$$

(c) The binding energy per nucleon for ${}^{4}_{2}He = \frac{28.4}{4} = 7.10$ MeV per nucleon

Example 3

(a) A nucleus has a nucleon number A, a proton number Z and a binding energy E_{be}.

The rest masses of the proton, neutron and mass of nucleus are m_p, m_n and $m_{nucleus}$, respectively, and c is the speed of light.

(i) Show that the mass of the nucleus $m_{nucleus}$ is given by

$$m_{nucleus} = (A - Z)m_n + Zm_p - \frac{E_{be}}{c^2}$$

(ii) If the nucleus is deuterium, represented by the symbol ${}^2_1H$, show that the mass of deuterium atom ($m_{deuterium\ atom}$) is given by

$$m_{deuterium\ atom} = m_n + m_{hydrogen\ atom} - \frac{E_{be}}{c^2}$$

where $m_{hydrogen\ atom}$ is the mass of hydrogen atom.

(b) Calculate in MeV, the binding energy per nucleon of deuterium from the information given below:

Atomic mass of deuterium $m_{deuterium\ atom}$ = 2.01410 u;
Atomic mass of hydrogen $m_{hydrogen\ atom}$ = 1.00783 u;
Rest mass of neutron m_n = 1.00867 u.

Solution 3

(a) (i) Binding energy $E_{be} = [(A - Z)m_n + Zm_p - m_{nucleus}] \times c^2$

$$\therefore m_{nucleus} = (A - Z)m_n + Zm_p - \frac{E_{be}}{c^2}$$

(ii) Binding energy of deuterium is given by

$$E_{be} = [(2 - 1)m_n + (1)m_p - m_{deuterium\ nucleus}] \times c^2$$

$$= [(1)m_n + (1)m_p - (m_{deuterium\ atom} - (1)m_e)] \times c^2$$

where m_e is the mass of an electron

$$= [m_n + m_p + m_e - m_{deuterium\ atom}] \times c^2$$

$$= [m_n + (m_p + m_e) - m_{deuterium\ atom}] \times c^2$$

since $m_{hydrogen\ atom} = m_p + m_e$

$$= [m_n + m_{hydrogen\ atom} - m_{deuterium\ atom}] \times c^2$$

$$\therefore m_{deuterium\ atom} = m_n + m_{hydrogen\ atom} - \frac{E_{be}}{c^2}$$

(b) Binding energy of deuterium is given by

$$E_{be} = [m_n + m_{hydrogen} - m_{deuterium}] \times c^2$$

$$= (1.00867\ u + 1.00783\ u - 2.01410\ u)\ (3.00 \times 10^8)^2$$

$$= (0.0024)(1.66 \times 10^{-27})(3.00 \times 10^8)^2$$

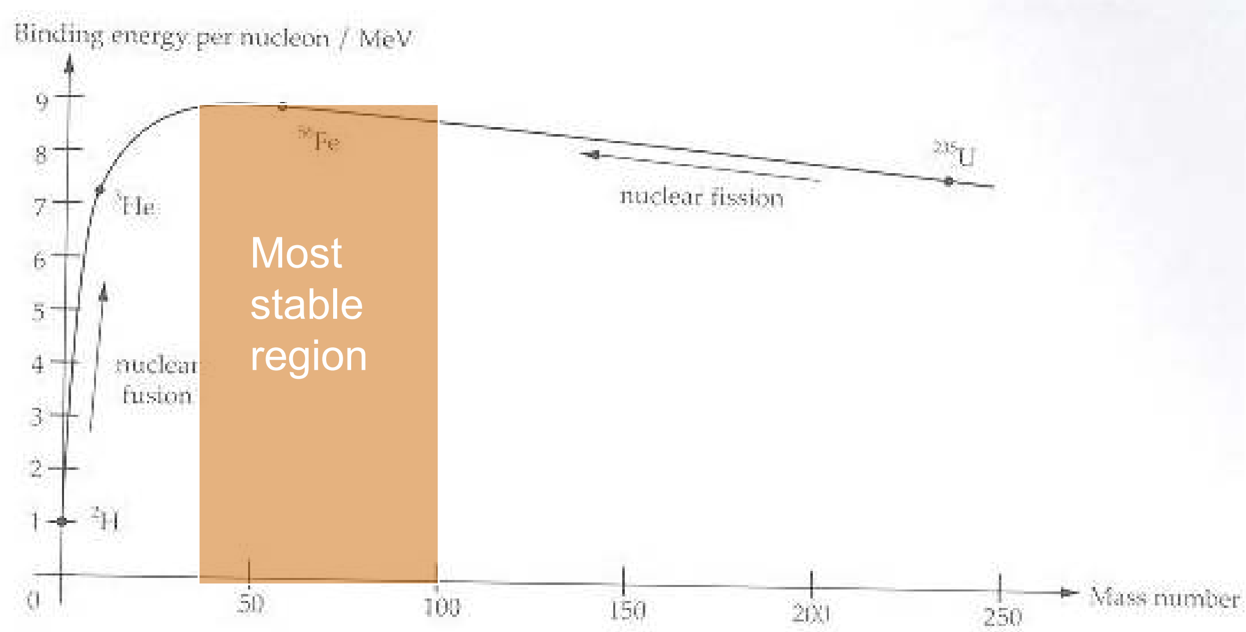
$$= 3.59 \times 10^{-13}\ J$$

Binding energy per nucleon of deuterium $= \dfrac{3.59 \times 10^{-13}}{2}$

$$= 1.79 \times 10^{-13}\ J$$

$$= 1.12\ MeV$$

Stability of Nuclei



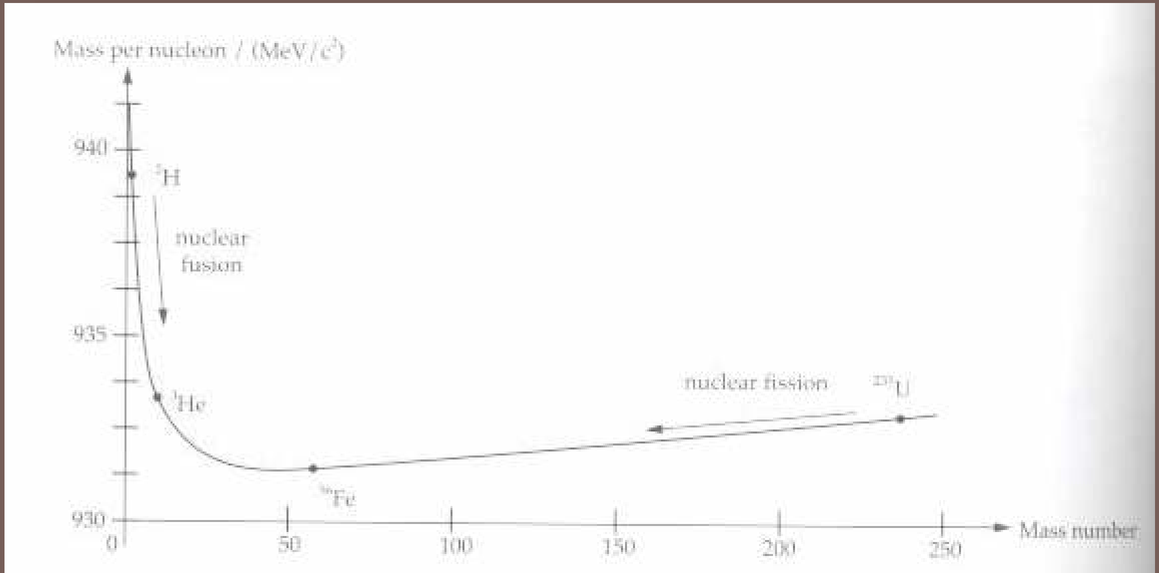
▲ Fig 20.5 Variation of binding energy per nucleon with mass number

The nucleus is more stable if it has a higher binding energy per nucleon. It would be more difficult to break up the nucleus as more energy is required to separate the nucleons.

The most stable nuclide can be found at the peak of the curve. It corresponds to the element $^{56}_{26}Fe$

It has the greatest mass defect and the highest binding energy per nucleon.

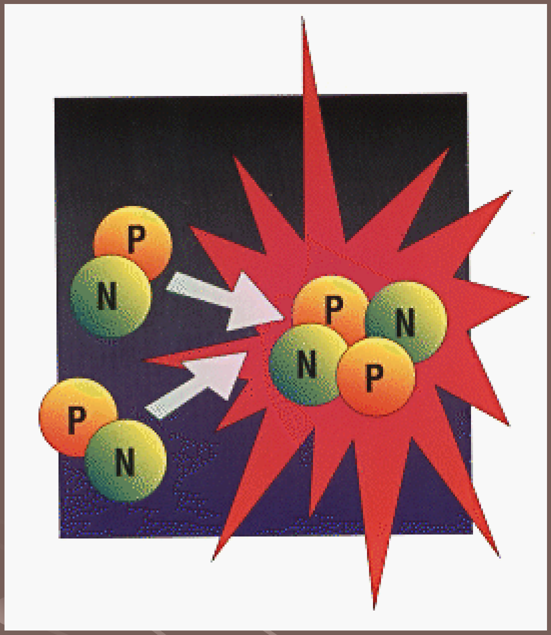
Mass per Nucleon



▲ Fig 20.6 Variation of mass per nucleon with mass number

- Mass per nucleon is small when binding energy per nucleon is high.
- Elements with very small or very large mass number are unstable.
- To attain stability
  - Nuclei with low mass numbers may undergo nuclear fusion
  - Nuclei with high mass numbers may undergo nuclear fission.

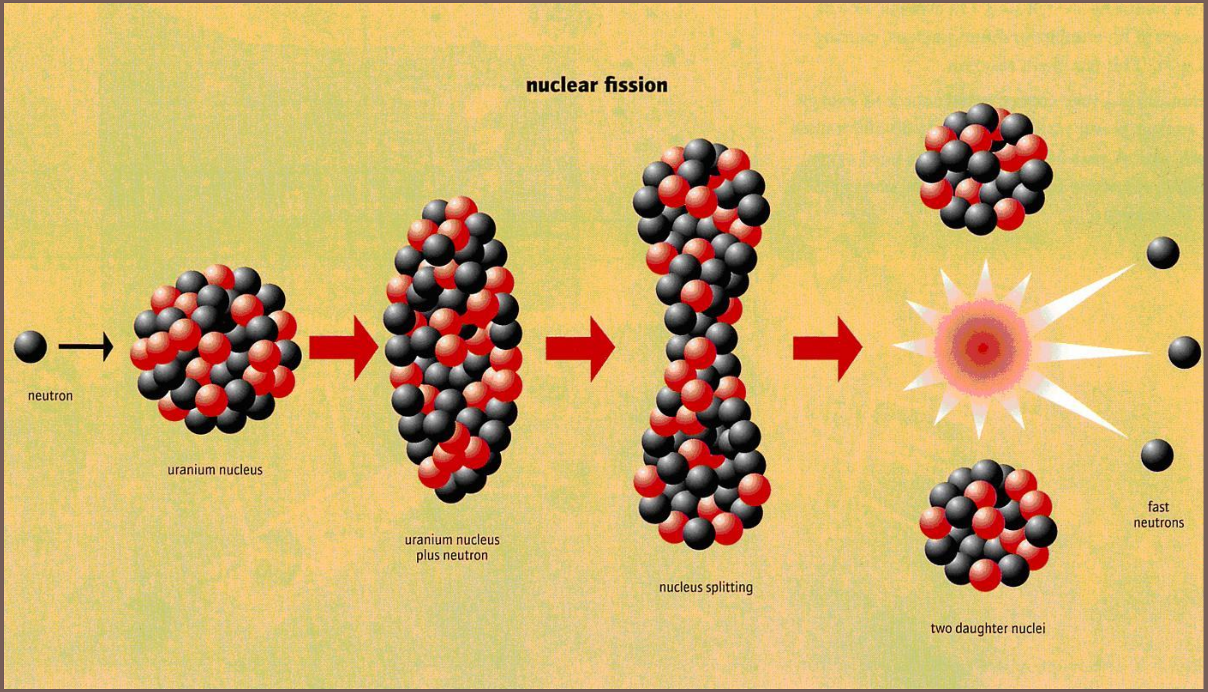
Nuclear Fusion



- Nuclei with low mass numbers may undergo nuclear fusion under certain conditions.
- In general nuclear fusion is possible as long as the final product has more binding energy per nucleon (i.e. less mass) than the reactants.
- The enormous amount of energy generated in the Sun is due to this process.
- Energy released in the fusion process is very much greater than energy released in the fission process
- An example of nuclear fusion:
  - two deuterium atoms fuse together to form helium-3 under extremely high temperature.
  - He-3 has a greater binding energy per nucleon and is more stable than deuterium.

$$^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} + \text{energy}$$

Nuclear Fission



- In general, heavier nuclides tend to disintegrate into lighter, more stable nucleus.
- Fission fragments have a greater binding energy per nucleon (i.e. less mass per nucleon) than the original nuclide.
- Example: Uranium-235 may absorbs a slow thermal neutron and splits into two part, Xenon-144 and Strontium-90.

$$ {}^{1}_{0}n + {}^{235}_{92}U \rightarrow {}^{236}_{92}U^{*} \rightarrow {}^{144}_{54}Xe + {}^{90}_{38}Sr + 2\,{}^{1}_{0}n $$

Example 4

One nuclear reaction that could occur when nucleus absorbs a slow neutron is:

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{236}_{92}U \rightarrow ^{146}_{57}La + ^{87}_{35}Br + 3X + Y$$

The binding energy per nucleon for each of the products is given by:

Binding energy per nucleon of $^{236}_{92}U$ = 7.59 MeV

Binding energy per nucleon of $^{146}_{57}La$ = 8.41 MeV

Binding energy per nucleon of $^{87}_{35}Br$ = 8.59 MeV

(a) (i) Identify X and Y and complete the equation.

(ii) Name this type of nuclear reaction.

(iii) Explain why neutrons are used to split uranium-235.

(b) Estimate the amount of energy released.

(c) Suggest how energy is released during this nuclear reaction.

Solution 4

(a) (i) X is a neutron and Y is γ-photon.

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{236}_{92}U \rightarrow ^{146}_{57}La + ^{87}_{35}Br + 3^{1}_{0}n + \gamma-\text{photon}$$

(ii) Nuclear fission reaction

(iii) Neutrons are used because they are electrically neutral and will be able to penetrate into the uranium nucleus without experiencing repulsion.

(b)

$$\begin{aligned}
\text{Binding energy of } ^{146}_{57}La \ E_{La} &= 146 \times 8.41 \\
&= 1227.86 \text{ MeV} \\
\text{Binding energy of } ^{87}_{35}Br \ E_{Br} &= 87 \times 8.59 \\
&= 747.33 \text{ MeV} \\
\text{Total binding energy of products} &= 1227.86 + 747.33 \text{ MeV} \\
&= 1975.19 \text{ MeV} \\
\text{Binding energy of } ^{236}_{92}U \ E_{U} &= 236 \times 7.59 \\
&= 1791.24 \text{ MeV} \\
\text{Amount of energy released} &= E_{La} + E_{Br} - E_{U} \\
&= 1975.19 - 1791.24 \\
&= 183.95 \text{ MeV} \\
&= 184 \text{ MeV}
\end{aligned}$$

(c) Energy is released as energetic γ-photons and the kinetic energies of $^{146}_{57}La$, $^{87}_{35}Br$ and neutrons. The latter appears as heat.

Example 5

1 kg of deuterium nuclides undergoes nuclear fusion only at very high temperatures of a few million degrees. The collision of deuterium nuclides is shown by the following reaction:

$$ {}^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{2}He + {}^{1}_{0}n $$

(a) Explain why nuclear fusion occurs only at very high temperatures.

(b) Suggest a method to attain such high temperatures on the Earth's surface.

(c) Calculate the amount of energy released by 1 kg of deuterium nuclides.

(d) A factory uses 10 000 kWh of electrical energy per month. Calculate the approximate number of fusion reactions that is required to keep the factory running for 1 year.

[Rest mass of ${}^{2}_{1}H$ = 2.014102 u, rest mass of ${}^{3}_{2}He$ = 3.016029 u, rest mass of neutron = 1.008665 u]

Solution 5

(a) At high temperatures, the fast moving deuterium nuclides will be able to overcome the strong electrostatic repulsion between them and approach each other. When they are close enough, fusion occurs and energy is released. At lower temperatures, deuterium nuclides have lower speed. They are repelled away from each other and are not able to come close enough for fusion to occur.

(b) On the Earth's surface, vast amount of heat can be generated in an atomic bomb explosion. High temperatures of millions of degrees from the fission reaction will activate the fusion process.

(c) ${}^2_1H + {}^2_1H \rightarrow {}^3_2He + {}^1_0n$

Total rest mass of two ${}^2_1H$ $= 2 \times 2.014102$ u
$= 4.028204$ u

Total mass of products $= 3.016029$ u $+ 1.008665$ u
$= 4.024694$ u

Energy released during reaction $= \Delta mc^2$
$= (4.028204 \text{ u} - 4.024694 \text{ u})(3.00 \times 10^8)^2$
$= (0.003510)(1.66 \times 10^{-27})(3.00 \times 10^8)^2$
$= 5.24 \times 10^{-13}$ J

Energy released per deuterium $= \dfrac{5.24 \times 10^{-13}}{2}$
$= 2.62 \times 10^{-13}$ J

In 0.002 kg, there are 6.02×10^{23} nuclei.
In 0.001 kg, there are 3.01×10^{23} nuclei.

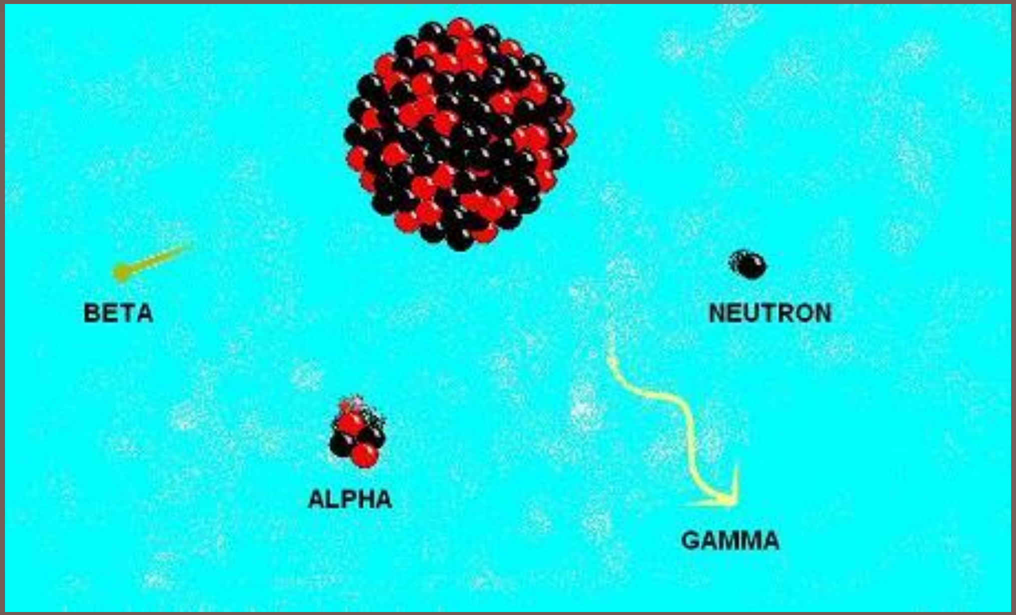
In 1 kg, there are $\dfrac{3.01 \times 10^{23}}{0.001} = 3.01 \times 10^{26}$ nucleus

$\therefore$ Energy released in 1 kg of deuterium $= 2.62 \times 10^{-13} \times 3.01 \times 10^{26}$
$= 7.89 \times 10^{13}$ J

(d) Energy required in 1 year $= 10000 \times 10^3 \times 3600 \times 12$
$= 4.32 \times 10^{11}$ J

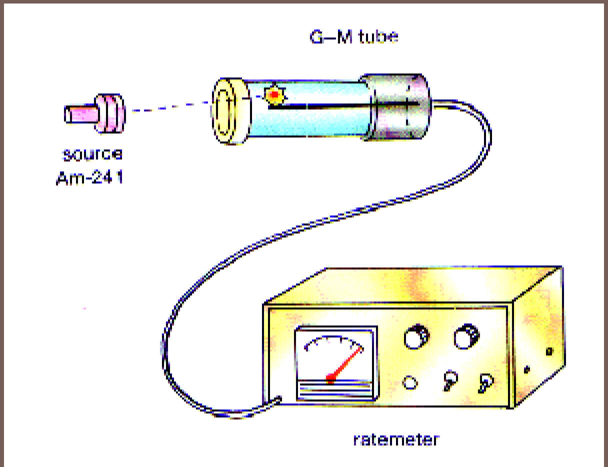
Number of reactions required $= \dfrac{4.32 \times 10^{11}}{5.24 \times 10^{-13}}$
$= 8.24 \times 10^{23}$

Radioactive Decay



Radioactive decay is the spontaneous and random disintegration of heavy unstable nucleus into more stable products with lower total mass through the emission of radiation such as alpha-particles, beta-particles and gamma-rays.

Measuring Radioactivity



- Radioactivity decay can be measured with a Geiger-Muller tube connected to a ratemeter. The ratemeter measures the count rate of the radioactive decay

- A radiation detector can register a count rate of 20-50 count per second, even in the apparent absence of radioactive materials. This is know as the background count. The radiation comes from low intensity radiation from small quantities of radioisotopes found in the ground, atmosphere and cosmic rays arriving at the surface of the Earth.

Decay Constant

In a random process, the **rate of radioactive decay** **-d*N*/d*t*** of a radioactive sample is directly proportional to the number ***N*** of radioactive nuclei present. That is,

$$-\frac{\mathrm{d}N}{\mathrm{d}t} \propto N$$

$$\frac{\mathrm{d}N}{\mathrm{d}t} = -\lambda N$$

where

λ = radioactivity decay constant; ***t*** = time

The decay constant λ is the fraction of the total number of atoms that decay per unit time.

Its S.I. unit is s^{-1}

Decay Constant

The radioactive law **d*N*/d*t* = –λ*N*** can be rewritten and integrated as follows:

$$\frac{\mathrm{d}N}{\mathrm{d}t} = -\lambda N$$

$$\frac{\mathrm{d}N}{N} = -\lambda \mathrm{d}t$$

$$\int_{N_0}^{N} \frac{\mathrm{d}N}{N} = \int_0^t -\lambda t$$

$$[\ln N]_{N_0}^{N} = [-\lambda t]_0^t$$

$$\ln \frac{N}{N_0} = -\lambda t$$

$$N = N_0 e^{-\lambda t}$$

In general $x = x_0 e^{-\lambda t}$

where *x* could represent
(a) activity *A*
(b) number of undecayed particles *N*
(c) count rate *C* or
(d) mass of undecayed particles *m*

Activity

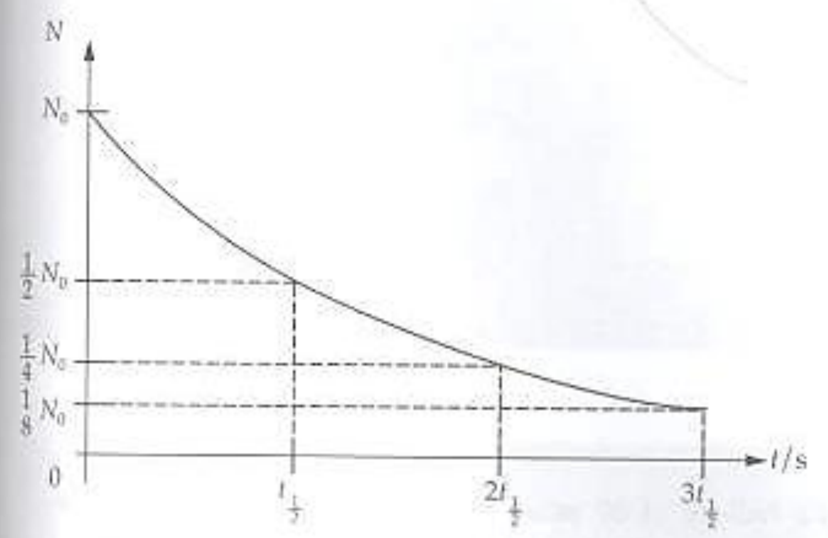
The activity A of a radioactive source is the number of disintegration it undergoes per unit time.

$$
\begin{aligned}
A &= -dN / dt \\
&= -\lambda N \\
&= -\lambda N_0 e^{-\lambda t} \\
&= A_0 e^{-\lambda t}
\end{aligned}
$$

Unit of A is becquerel, Bq

1 Bq = 1 decay s^{-1}

Graphical Representation



▲ Fig. 20.7 Radioactive decay graph

Time	Number of radioactive nuclei that remains	Number of radioactive nuclei that have disintegrated
0	$\left(\frac{1}{2}\right)^0 N_0 = N_0$	0
$1\,t_{\frac{1}{2}}$	$\left(\frac{1}{2}\right)^1 N_0 = \frac{1}{2}N_0$	$\frac{1}{2}N_0$
$2\,t_{\frac{1}{2}}$	$\left(\frac{1}{2}\right)^2 N_0 = \frac{1}{4}N_0$	$\frac{3}{4}N_0$
$3\,t_{\frac{1}{2}}$	$\left(\frac{1}{2}\right)^3 N_0 = \frac{1}{8}N_0$	$\frac{7}{8}N_0$

Half Life

The half-life of a radioactive nuclide is the time taken for the number of radioactive nuclide to disintegrate to half its initial value.

Using $N = N_0 e^{-\lambda t}$

when $t = t_{\frac{1}{2}}$, the number of undecayed atoms $N = \frac{N_0}{2}$.

Therefore, $\frac{N_0}{2} = N_0 e^{-\lambda t_{\frac{1}{2}}}$

$$2 = e^{\lambda t_{\frac{1}{2}}}$$

$$\ln 2 = \lambda t_{\frac{1}{2}}$$

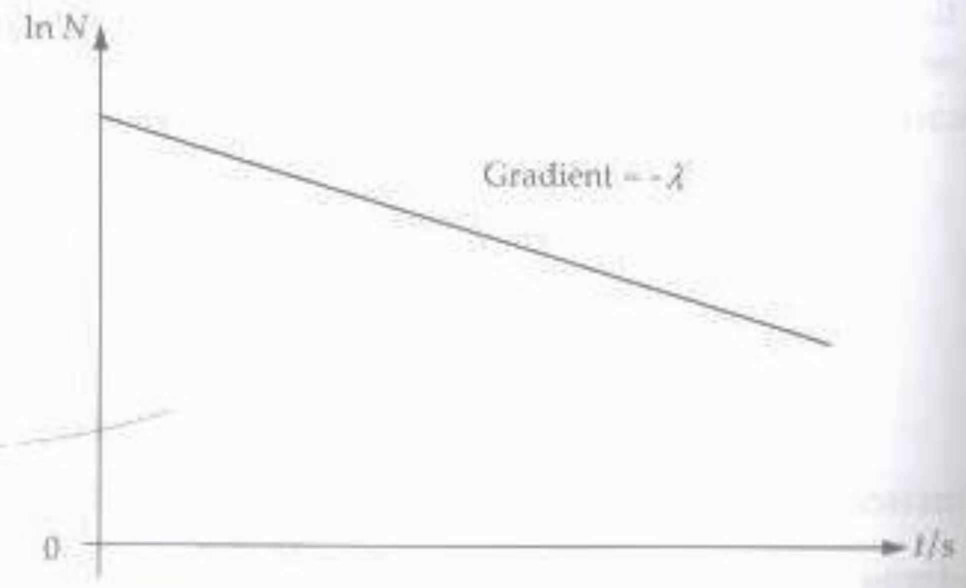
and so $t_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$ or $t_{\frac{1}{2}} = \frac{0.693}{\lambda}$

Graph of ln*N* against *t*

Since $N = N_0 e^{-\lambda t}$

$$\ln N = -\lambda t + \ln N_0$$

The graph of ln N against t is shown in Fig. 20.8.



Half Life of Some Materials

- uranium = 4500 million years
- radium = 1600 years
- polonium = 138 days
- radioactive lead = 27 minutes
- radon = 1 minute

Example 6

The half-life of radium is 1.62×10^3 years. What fraction of radium would remain after 1.0×10^3 years?

Solution

$$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$$

$$= \frac{\ln 2}{1.62 \times 10^3}$$

$$\lambda t = \left(\frac{\ln 2}{1.62 \times 10^3}\right)(1.0 \times 10^3)$$

$$= 0.617 \ln 2$$

Since $N = N_0 e^{-\lambda t}$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$= e^{-0.617 \ln 2}$$

$$= 0.652$$

Example 7

Radioactive caesium-137 has a half-life of 30 years. If the initial number of caesium nuclei is 10^4, what is the number of caesium nuclei that have decayed after 90 years?

Method 1:

90 years is equivalent to $\frac{90}{30} = 3$ half-lives

After 90 years, the number of caesium nuclei that remain is $\left(\frac{1}{2}\right)^3 \times 10\,000 = 1250$

After 90 years, the number of caesium nuclei that have decayed is
$10\,000 - 1250 = 8750$

Method 2:

As $t_{\frac{1}{2}} = 30$ years,

$$\lambda = \frac{\ln 2}{30 \times 365 \times 24 \times 3600}$$

$$\lambda t = \left(\frac{\ln 2}{30 \times 365 \times 24 \times 3600}\right)(90 \times 365 \times 24 \times 3600)$$

$$= 3 \ln 2$$

Since $N = N_o e^{-\lambda t}$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$= e^{-3 \ln 2}$$

$$= 0.125$$

After 90 years, the number of caesium nuclei that have decayed $= (1 - 0.125) \times 10\,000 = 8750$

Example 8

A high speed alpha particle collides with a stationary nitrogen nucleus $^{14}_{7}N$ to produce an unknown nucleus $^{A}_{Z}X$, a proton and a γ-photon.

$$^{14}_{7}N + ^{4}_{2}He \rightarrow ^{A}_{Z}X + ^{1}_{1}p + \gamma$$

(a) Identify the unknown nucleus X and state the number and type of particles which form this nucleus.

(b) Calculate the minimum kinetic energy of the alpha particle that will make this nuclear reaction possible.

(c) Calculate the minimum velocity of the alpha particle.

(d) Describe the transformation of the kinetic energy of the alpha particle after the reaction.

The masses of the nuclei involved are given below.

Mass of $^{14}_{7}N = 2.32530 \times 10^{-26}$ kg;

Mass of $^{17}_{8}O = 2.82282 \times 10^{-26}$ kg;

Mass of proton $= 0.16735 \times 10^{-26}$ kg;

Mass of alpha particle $= 0.66466 \times 10^{-26}$ kg.

Solution 8

(a) The unknown nucleus is oxygen ($^{17}_{8}O$).

$$\text{Number of protons} = 8$$
$$\text{Number of neutrons} = 17 - 8 = 9$$

(b) $$\text{Mass of } ^{14}_{7}N + \text{mass of } ^{4}_{2}He = (2.32530 \times 10^{-26}) + (0.66466 \times 10^{-26}) = 2.98996 \times 10^{-26}\text{ kg}$$

$$\text{Mass of } ^{17}_{8}O + \text{mass of } ^{1}_{1}p = (2.82282 \times 10^{-26}) + (0.16735 \times 10^{-26}) = 2.99017 \times 10^{-26}\text{ kg}$$

Since (mass of $^{14}_{7}N$ + mass of $^{4}_{2}He$) < (mass of $^{17}_{8}O$ + mass of $^{1}_{1}p$), energy must be supplied before the nuclear reaction can take place. This energy comes from the kinetic energy of the alpha particle.

Therefore, minimum energy required

$$= \Delta mc^2$$
$$= [(2.99017 \times 10^{-26}) - (2.98996 \times 10^{-26})] \times (3.0 \times 10^{8})^2$$
$$= 1.89 \times 10^{-13}\text{ J}$$

(c) Let the minimum velocity of the alpha particle be v.

$$\frac{1}{2}mv^2 = 1.89 \times 10^{-13}$$

$$v = \sqrt{\frac{2(1.89 \times 10^{-13})}{(0.66466 \times 10^{-26})}} = 7.54 \times 10^{6}\text{ m s}^{-1}$$

(d) The kinetic energy of the alpha particle is transformed into the kinetic energy of oxygen and hydrogen nucleus as well as the energy of γ-photon.

Example 9

A stationary radium nucleus may decay spontaneously into a radon nucleus and an α-particle as shown below.

$$^{226}_{88}\text{Ra} \rightarrow \, ^{222}_{86}\text{Rn} + \, ^{4}_{2}\text{He}$$

The rest masses of these nuclei are:

$$^{226}_{88}\text{Ra} = 226.0254 \text{ u}; \; ^{222}_{86}\text{Rn} = 222.0175 \text{ u}; \; ^{4}_{2}\text{He} = 4.0026 \text{ u}$$

(a) What physical quantities are conserved in this nuclear reaction?

(b) Calculate the binding energy of $^{226}_{88}\text{Ra}$.

(c) The total kinetic energy of the decay products is 4.78 MeV. Calculate the wavelength of the gamma photon that was emitted.

Solution 9

(a) The physical quantities that are conserved are:

(1) nucleon number

(2) proton number

(3) total linear momentum

(4) total mass-energy

(b) The rest mass of $^{226}_{88}Ra$ = 226.0254 u

The rest mass of $^{222}_{86}Rn$ = 222.0175 u

The rest mass of $^{4}_{2}He$ = 4.0026 u

Total mass of the products = 222.0175 u + 4.0026 u

= 226.0201 u

Binding Energy of $^{226}_{88}Ra$ = [226.0254 u − 226.0201 u] × c^2

$= (0.0053 \times 1.66 \times 10^{-27})(3.00 \times 10^8)^2$

$= 7.92 \times 10^{-13}$ J

= 4.95 MeV

(c) The energy of the photon = 4.95 − 4.78

= 0.17 MeV

Wavelength of gamma photon is given by

$$\lambda = \frac{hc}{E}$$

$$= \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{0.17 \times 10^6 \times 1.60 \times 10^{-19}}$$

$$= 7.3 \times 10^{-12} \text{ m}$$

26.4 The spontaneous and random nature of radioactive decay

Some elements have nuclei which are unstable. In order to become more stable, they emit particles and/or electromagnetic radiation (see AS Level Topic 26). The nuclei are said to be **radioactive**.

Detection of the count-rate of radioactive sources shows that the emission of radiation is both spontaneous and random.

Radioactive decay is a spontaneous process because it is not affected by any external factors, such as temperature or pressure.

Decay is random in that it is not possible to predict which nucleus in a sample will decay next.

Radioactive decay is a random process in that it cannot be predicted which nucleus will decay next. There is a constant probability that a nucleus will decay in any fixed period of time.

Random decay

We now look at some of the consequences of the random nature of radioactive decay. If six dice are thrown simultaneously (Figure 26.6), it is likely that one of them will show a six.

Figure 26.6 The dice experiment

If 12 dice are thrown, it is likely that two of them will show a six, and so on. While it is possible to predict the likely number of sixes that will be thrown, it is impossible to say which of the dice will actually show a six. We describe this situation by saying that the throwing of a six is a **random process**.

In an experiment similar to the one just described, some students throw a large number of dice (say 6000). Each time a six is thrown, that die is removed. The results for the number of dice remaining after each throw are shown in Table 26.1.

Table 26.1 Results of dice-throwing experiment

number of throws	number of dice remaining	number of dice removed
0	6000	
1	5000	1000
2	4173	827
3	3477	696
4	2897	580
5	2414	483
6	2012	402
7	1677	335
8	1397	280

Figure 26.7 is a graph of the number of dice remaining against the number of throws.

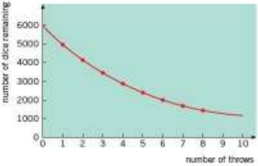


Figure 26.7

This kind of graph is called a **decay curve**. The rate at which dice are removed is not linear, but there is a pattern. After between 3 and 4 throws, the number of dice remaining has halved. Reading values from the graph shows that approximately 3.8 throws would be required to halve the number of dice. After another 3.8 throws the number has halved again, and so on.

We can apply the dice experiment to model radioactive decay. The 6000 dice represent radioactive nuclei. To score a six represents radioactive emission. All dice scoring six are removed, because once a nucleus has undergone radioactive decay, it is no longer available for further decay. Thus, we can describe how rapidly a sample of radioactive material will decay.

A graph of the number of undecayed nuclei in a sample against time has the typical decay curve shape shown in Figure 26.8.

It is not possible to state how long the entire sample will take to decay (its 'life'). However, after 3 minutes the number of undecayed nuclei in the sample has halved. After a further 3 minutes, the number of undecayed atoms has halved again. We describe this situation by saying that this radioactive isotope has a **half-life** $t_{\frac{1}{2}}$ of 3 minutes.

The half-life of a radioactive nuclide is the time taken for the number of undecayed nuclei to be reduced to half its original number.

The half-lives of different isotopes have a very wide range of values. Examples of some radioactive isotopes and their half-lives are given in Table 26.2.

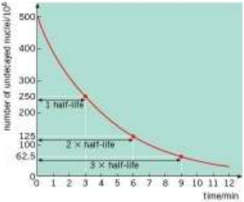


Figure 26.8 Radioactive decay curve

Table 26.2 Examples of half-life

radioactive isotope	half-life
uranium-238	4.5×10^9 years
radium-226	1.6×10^3 years
radon-222	3.8 days
francium-221	4.8 minutes
astatine-217	0.03 seconds

We shall see later that half-life may also be expressed in terms of the activity of the material.

If you measure the count rate from an isotope with a very long half-life, you might expect to obtain a constant value. In fact, the count rate fluctuates about an average value. This demonstrates the random nature of radioactive decay.

Example

The half-life of francium-221 is 4.8 minutes. Calculate the fraction of a sample of francium-221 remaining undecayed after a time of 14.4 minutes.

The half-life of francium-221 is 4.8 min, so after 4.8 min half of the sample will remain undecayed. After two half-lives (9.6 min), $0.5 \times 0.5 = 0.25$ of the sample will remain undecayed. After three half-lives (14.4 min), $0.5 \times 0.25 = 0.125$ will remain undecayed. So the fraction remaining undecayed is **0.125** or **1/8**.

Now it's your turn

4 Using the half-life values given in Table 26.2, calculate:
  (a) the fraction of a sample of uranium-238 remaining undecayed after 9.0×10^9 years,
  (b) the fraction of a sample of astatine-217 remaining undecayed after 0.30 s,
  (c) the fraction of a sample of radium-226 that has decayed after 3200 years,
  (d) the fraction of a sample of radon-222 that has decayed after 15.2 days.

In carrying out experiments with radioactive sources, it is important to take account of background radiation. In order to determine the count-rate due to the radioactive source, the background count-rate must be subtracted from the total measured count-rate. Allowance for background radiation gives the corrected count-rate.

Physics is Great

Enjoy Your Study!